



MC321/IE406—Machine Learning (3-0-2-4)

AY:2025-2026 (Winter)

Course Placement: Machine learning is B.Tech MnC Core course in the 6th semester, and an ICT elective course for B.Tech-ICT, ICT+CS 3rd year (6th Semester)

Course Format: It is **3 hours' lecture** and **2 hours' lab** every week.

Instructor: Srimanta Mandal (srimanta_mandal@dau.ac.in)

Course Objective:

The objective of this course is to introduce different classical techniques for machine learning such that the students can understand the working principle of different techniques. The course shall also help the students to address different real life problems with machine learning techniques. Moreover, the project part of the course is completely devised to encourage student to explore the area of machine learning by research and development.

Pre-requisites:

- i) Probability, Random Variable
- ii) Linear Algebra
- iii) Optimization
- iv) Programming (Python—preferable /MATLAB)

Course Content:

- 1) *Regression* – Linear regression, Convexity and gradient descent algorithm, Vectorization, Feature Scaling, Concept of Regularization, Normal Equation, Probabilistic Linear Regression, Overfitting, Ridge regression, The Lasso, Basis Functions
- 2) *Resampling, Model Selection*– Generalization Error, Model Capacity, Cross-validation, The Bootstrap, Bias-Variance Decomposition
- 3) *Classification* – Distance Based Classification, K-NN, Naïve Bayes Classifier, Perceptron, Logistic regression, Softmax, Decision trees, Boosting, Random forest, Support Vector Machine
- 4) *Clustering* – K-Means, Gaussian Mixture Model
- 5) *Optimal Projection of Data* – Principal component analysis, Linear Discriminant Analysis
- 6) *Neural Network*- Representation, Forward propagation, Back Propagation

Readings:

1. Machine Learning: A probabilistic Perspective by Kevin Murphy
2. Pattern Recognition and Machine Learning by Christopher Bishop
3. Elements of Statistical Learning by Trevor and Hastie
4. Research Articles (suggested by the instructors)

Assessment Method (Tentative):

- 1st In-semester examination – 20%
- 2nd In-semester examination – 25%
- End-semester examination – 35%
- Lab assignments – 20%

Course Outcomes:

After completion of this course, students should be able to:

- Understand the fundamental machine learning techniques. [P1, P12]
- Analyze a problem, and formulate a model to address the same. [P2, P3, P4]
- Analyze the shortcomings of existing methods, and ponder upon those. [P2, P5]
- Work in a group for assignments, and present the results [P9, P10].

Mapping of Course Outcomes to Program Outcomes

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X	X				X	X		X

Lecture Schedule (Tentative):

Sl. No.	Description		No. of Lectures
1	Regression		11
	1.1	Linear regression	
	1.2	Convexity and gradient descent algorithm	
	1.3	Vectorization	
	1.4	Feature Scaling, The concept of Regularization, Normal Equation	
	1.5	Probabilistic Linear Regression	

	1.6	Ridge regression, The Lasso	
	1.7	Basis Functions	
2	Resampling-Model Selection		4
	2.1	Generalization Error	
	3.2	Model Capacity	
	3.3	Cross-validation	
	3.4	The Bootstrap	
	3.5	Bias-Variance Decomposition	
3	Classification		14
	3.1	Distance Based Classification	
	3.2	K-NN, Density Estimation	
	3.3	Naïve Bayes Classifier	
	3.4	Perceptron	
	3.5	Logistic Regression, Newton's Method, Exponential Family	
	3.6	Softmax	
	3.7	Decision Tree, Ensemble Methods	
	3.8	Support Vector machine and Kernel	
4	Clustering		4
	4.1	K-means	
	4.2	Gaussian Mixture Model	
5	Optimal Projection of Data		4
	5.1	Principal Component Analysis	
	5.2	Linear Discriminant Analysis	
6	Neural Networks		5
	6.1	Representation	
	6.2	Forward Propagation	
	6.3	Backpropagation	